NY ML RESEARCH GUILD · AMAZON JFK27 · OCT 14

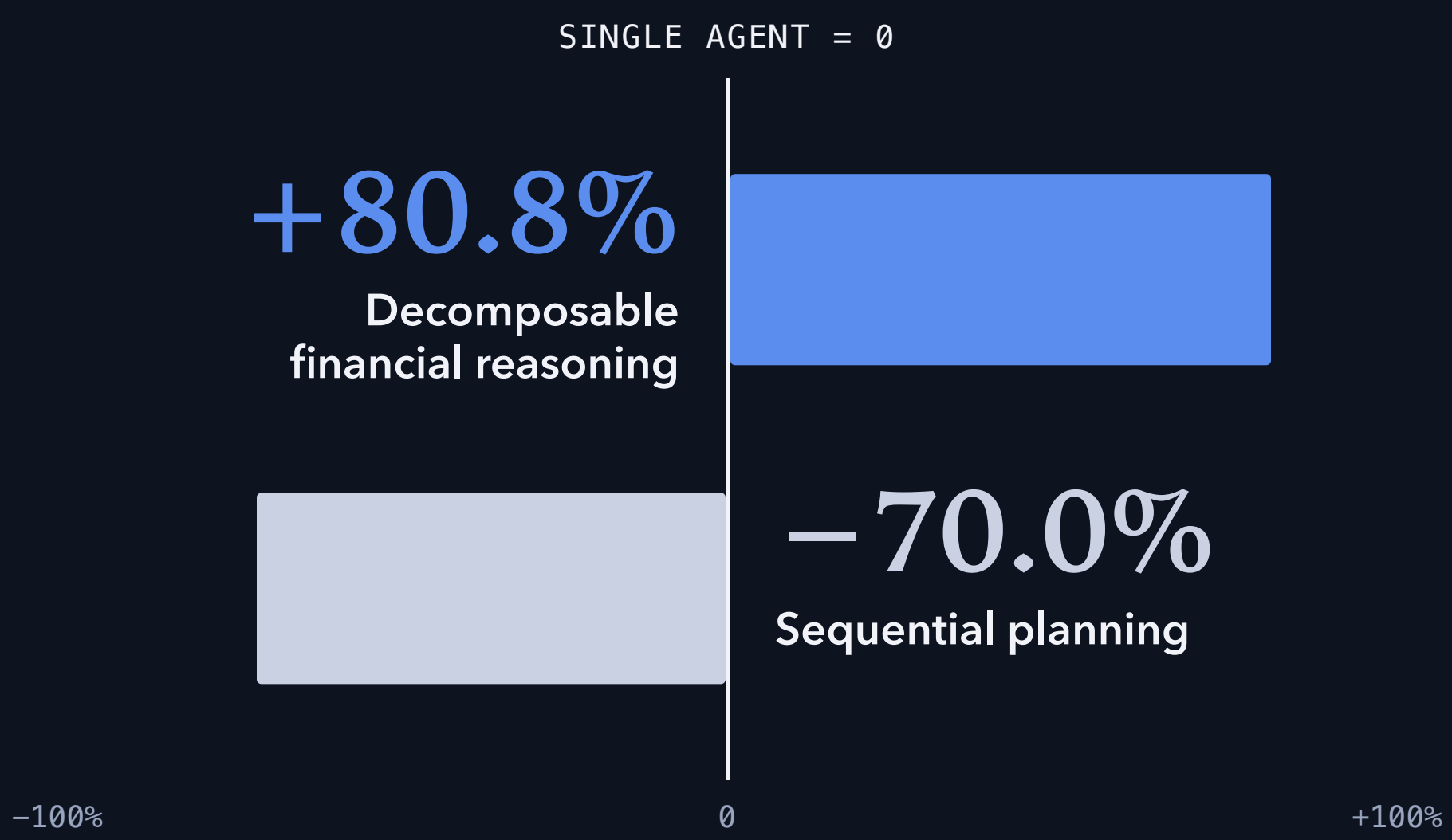
More agents, or more compute?

From chain of thought to agent swarms. When a swarm wins, what actually did the work?

Alex Yedi

MULTI-AGENT VS ONE AGENT

Same idea, opposite results



Independent pieces: the team tended to help. **Chained steps: it tended to hurt.**

Source: Kim et al., "Towards a Science of Scaling Agent Systems," arXiv 2512.08296, 2025

THE FAMOUS SWARM RESULT

More findings. More tokens, too.

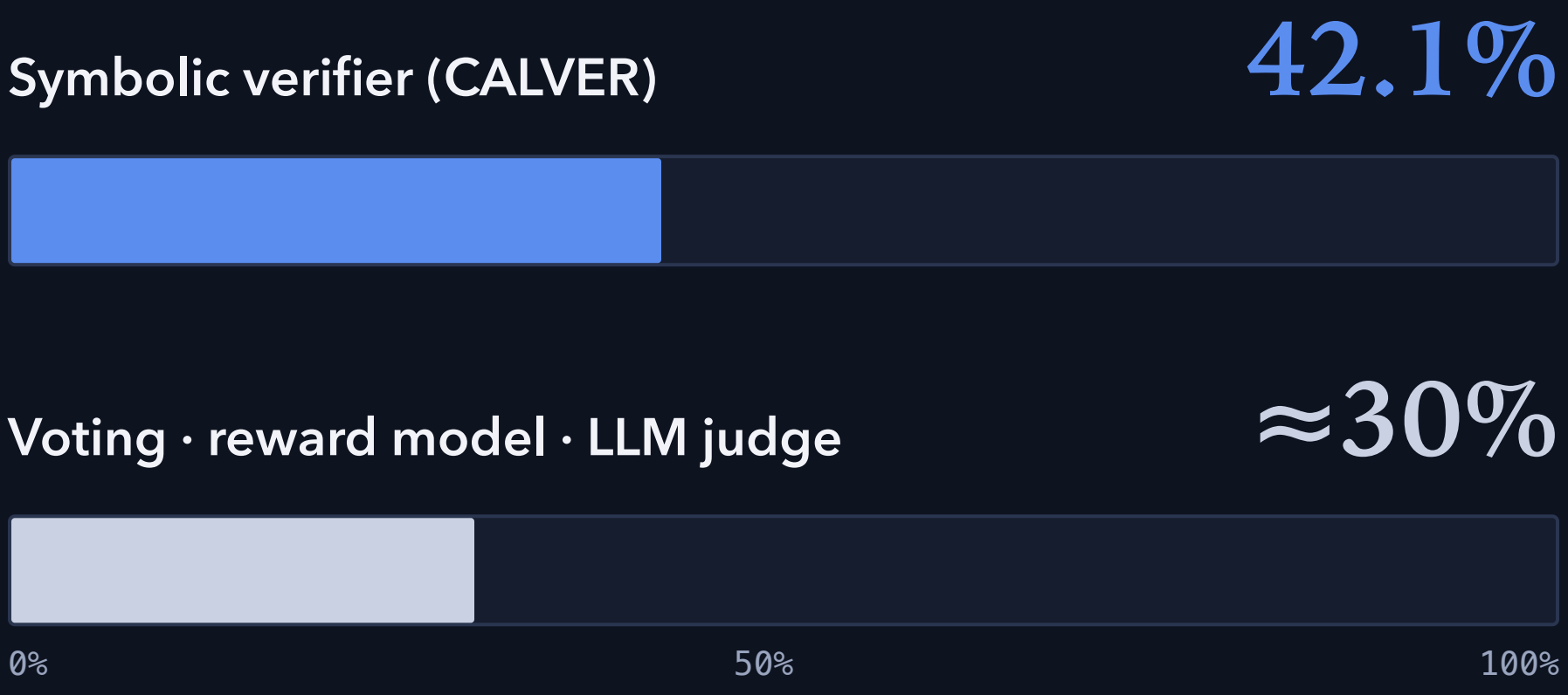


Not compute-matched. The swarm also searched a wider scope. In shared scope, cost per finding was roughly comparable.

Source: Anthropic, "Multiagent Systems: Patterns and Problems," 2026

WHEN MANY ANSWERS ARE VALID

Checking beats counting votes



A swarm hands back a pile of attempts. **Agreement isn't proof when several answers are right.**

Source: Vaidya, Jerzak, Sprague, Yin, Ho, CALVER, arXiv 2608.03506, 2026 (CLEAR find-one-valid task)

BEFORE YOU BELIEVE A SWARM RESULT

Ask for the baseline

01 What does one agent do with the same tokens?

02 Does the task split into independent pieces?

03 What checks the answers: votes or a verifier?

Separate agents, extra samples, or the checker?

Alex Yedi